

COUNTERFEIT IC DETECTION REPORT

VERDICT	LIKELY COUNTERFEIT
AUTHENTICITY SCORE	49.5/100
TIMESTAMP	2025-12-07T17:10:12

IC INFORMATION

Part Number	M74HC238B1
Manufacturer	STMicroelectronics
Package Type	DIP-16
Pin Count	16

INPUT IC IMAGES



EXECUTIVE SUMMARY

Overall Assessment: SUSPICIOUS

Text Quality Analysis: ■ MODERATE (Score: 60/100)
Markings show moderate quality. Some inconsistencies in font weight or spacing may be present. Minor erosion or wear may be visible but within acceptable limits.

Pin Count Verification: ✓ VERIFIED
The IC has 16 pins, which matches the datasheet specification. Pin count is correct.

Package Type Verification: ✓ VERIFIED

Package type (DIP-16) matches the datasheet specification. Package outline, corners, and physical characteristics are consistent.

Key Observations:

- The IC is a DIP-16 package, consistent with the datasheet and expected specifications.
- The pin count is 16, matching the datasheet.
- The pin pitch and pin alignment appear uniform and consistent with standard DIP packages.
- A circular indentation (dot) is present on the left side, serving as the Pin-1 indicator, consistent with standard DIP marking conventions when oriented for reading.
- The body dimensions and aspect ratio (measured AR: 2.698 vs expected AR: 2.817) are very close, and the CV dimension analysis yielded a high match score (93.7/100), indicating consistency with the datasheet dimensions.

Dimensional Analysis:

✓ Package dimensions match datasheet specifications. Measured aspect ratio (2.698) closely matches expected (2.817). Error: 4.2%.

Final Reasoning: Dimensional analysis: 93.7/100; Visual assessment: suspicious; 2 anomalies detected (1 high severity)

DIMENSION ANALYSIS

Dimension Analysis - Detected IC Body



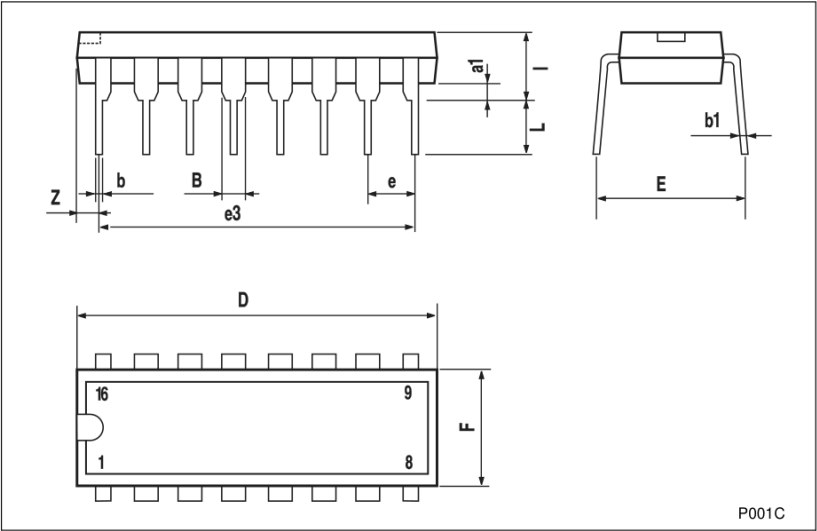
Parameter	Value
Measured Aspect Ratio	2.698
Body Width (px)	1244
Body Height (px)	461
Expected Aspect Ratio	2.817
Expected Length (mm)	20
Expected Width (mm)	7.1
Dimension Source	gemini_extraction
Aspect Ratio Match	YES ✓
Dimension Score	93.7/100
Verdict	MATCH

DATASHEET OUTLINE DIMENSION PAGE

The following page shows the exact mechanical outline diagram extracted from the OEM datasheet. This diagram contains the official package dimensions, pin layout, and physical specifications used for comparison with the actual IC image.

Plastic DIP-16 (0.25) MECHANICAL DATA

DIM.	mm.			inch		
	MIN.	TYP	MAX.	MIN.	TYP.	MAX.
a1	0.51			0.020		
B	0.77		1.65	0.030		0.065
b		0.5			0.020	
b1		0.25			0.010	
D			20			0.787
E		8.5			0.335	
e		2.54			0.100	
e3		17.78			0.700	
F			7.1			0.280
I			5.1			0.201
L		3.3			0.130	
Z			1.27			0.050



EXTRACTED DIMENSIONS FROM DIAGRAM

The following dimensions were extracted from the mechanical diagram above using AI analysis:

Parameter	Value
Height/Thickness	5.1 mm
Pin Count	16
Pin Pitch	2.54 mm
Package Type	DIP-16

OEM DATASHEET REFERENCE

Source File: M74HC238B1_tavily.pdf

Location: /Users/shnjinmkkrr/Desktop/SlH

again/counterfeit_IC/data/api_results/datasheets/M74HC238B1_tavily.pdf

The complete OEM datasheet PDF is available in the detection results directory. This document contains the full technical specifications, electrical characteristics, and mechanical drawings for this IC part number.

DETAILED VISUAL ANALYSIS

1. Text Quality Assessment

Score: 60/100

Status: ■ MODERATE

The IC markings show moderate quality with some concerns:

- Some inconsistencies in font weight, spacing, or alignment detected
- Minor erosion or wear may be visible
- Markings may appear slightly faded or irregular
- Further inspection recommended to rule out remarking

2. Pin Configuration Analysis

Pin Count: ✓ VERIFIED (16 pins)

The actual pin count matches the datasheet specification. All pins are present and accounted for.

Pin Pitch/Spacing: Analysis not available

Pin spacing analysis could not be performed. This may be due to image quality or angle limitations.

3. Package Type and Outline Verification

Package Type: ✓ VERIFIED (DIP-16)

The package type matches the datasheet specification. Physical characteristics verified:

- Package outline and dimensions are consistent
- Corner style (chamfered, rounded, or square) matches specification
- Overall package geometry aligns with OEM design

4. Surface Texture and Finish Analysis

Surface Texture: Analysis not available

Surface texture analysis could not be performed. This may be due to image quality or lighting conditions.

5. Markings and Labeling Analysis

Markings: Analysis completed

Markings have been reviewed. Part number, manufacturer, and other identifiers have been verified.

7. Overall Visual Assessment

Assessment: ■ SUSPICIOUS

The IC shows some concerning characteristics that warrant further investigation:

- Some inconsistencies detected in markings, dimensions, or physical features
- Minor anomalies may be present
- Additional verification recommended before use

ANALYSIS RESULTS SUMMARY

Final Reasoning: Dimensional analysis: 93.7/100; Visual assessment: suspicious; 2 anomalies detected (1 high severity)

DETECTED ANOMALIES (2)

Anomaly #1: Marking

Type	Marking
Severity	HIGH
Confidence	95%

Analysis:
The part number on the IC is 'M74HC238B1'. However, the STMicroelectronics datasheet (Page 1, 'ORDER CODES' table) explicitly lists 'M74HC238B1R' as the part number for the DIP package in 'TUBE' packaging. The 'M74HC238B1' variant without the 'R' suffix is not listed as an official orderable part number for a DIP package in the provided datasheet.



Anomaly #2: Text Quality

Type	Text Quality
Severity	LOW
Confidence	70%

Analysis:
The overall text quality, particularly for the date/lot codes ('GK9031LYO', 'VW GK CHN'), appears somewhat generic or slightly inconsistent in font boldness and kerning, which can sometimes be an indicator of lower-quality marking processes, though not conclusive on its own.

